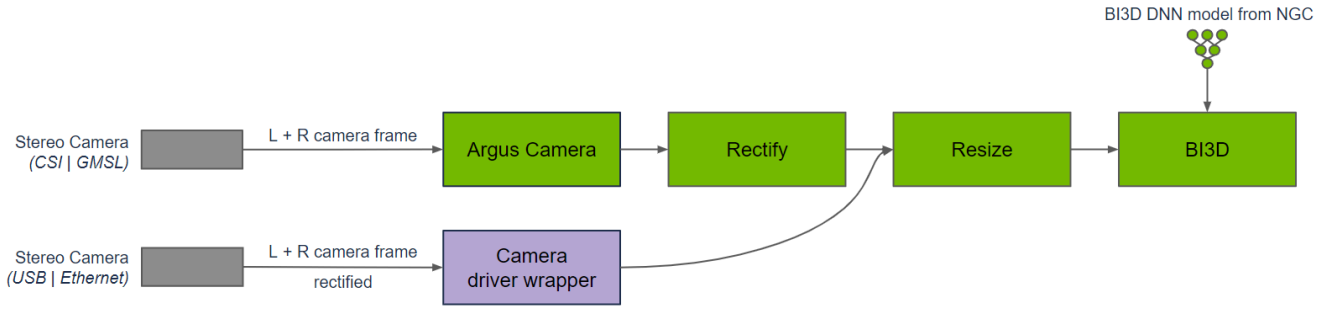
Isaac ROS Deep segmentation

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- Overview
- Quick experience

Isaac ROS Depth Segmentation official website link: https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_depth_segmentation/index.html

Overview

Isaac ROS Depth Segmentation provides NVIDIA-accelerated depth segmentation packages. Bi3D uses an optimized DNN model to perform binary depth estimation on stereo images, which can determine whether an obstacle is in a nearby area and estimate the free space of the ground.



Quick experience

First start the container with the following command

```
./run_isaac.sh
```

Then confirm the container on the host with `docker ps`, and open four container terminals with `docker exec -it <container name> bash`. Each terminal has to use the same ROS/FastDDS environment, for example:

```
Jetson Orin Nano L4T R39.2.1 · (8GB + 8GB ZRAM)
RDP 192.168.2.61:3389 jetson/yahboom [LAN]
VNC 192.168.2.61:5900 pass: yahboom [LAN]
jetson@jetson:~$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED
      STATUS          PORTS     NAMES
8f397e4b6c3e   my-isaac-ros-slam:4.6-all   "/bin/bash -c '\ncat …"   42 seconds
ago   Up 41 seconds             magical_bhabha
jetson@jetson:~$
```

Note: all the terminals below are executed inside the container:

Terminal A: start the Nano pool=8 ESS. The input image is 1920×1080 and the model input is 960×576:

```
ros2 launch /workspaces/isaac_ros-dev/launch/ess_nano_pool8.launch.py \
  engine_file_path:=/workspaces/isaac_ros-
dev/isaac_ros_assets/models_pending/dnn_stereo_disparity/dnn_stereo_disparity_v4.1.0_onnx_tr
t10.16/ess.engine \
  ess_plugin_file_path:=/workspaces/isaac_ros-
dev/isaac_ros_assets/models_pending/dnn_stereo_disparity/dnn_stereo_disparity_v4.1.0_onnx_tr
t10.16/plugins/aarch64/ess_plugins.so \
  input_image_width:=1920 input_image_height:=1080 \
  model_input_width:=960 model_input_height:=576
```

Terminal B: play the input in a loop at 0.25 times the speed:

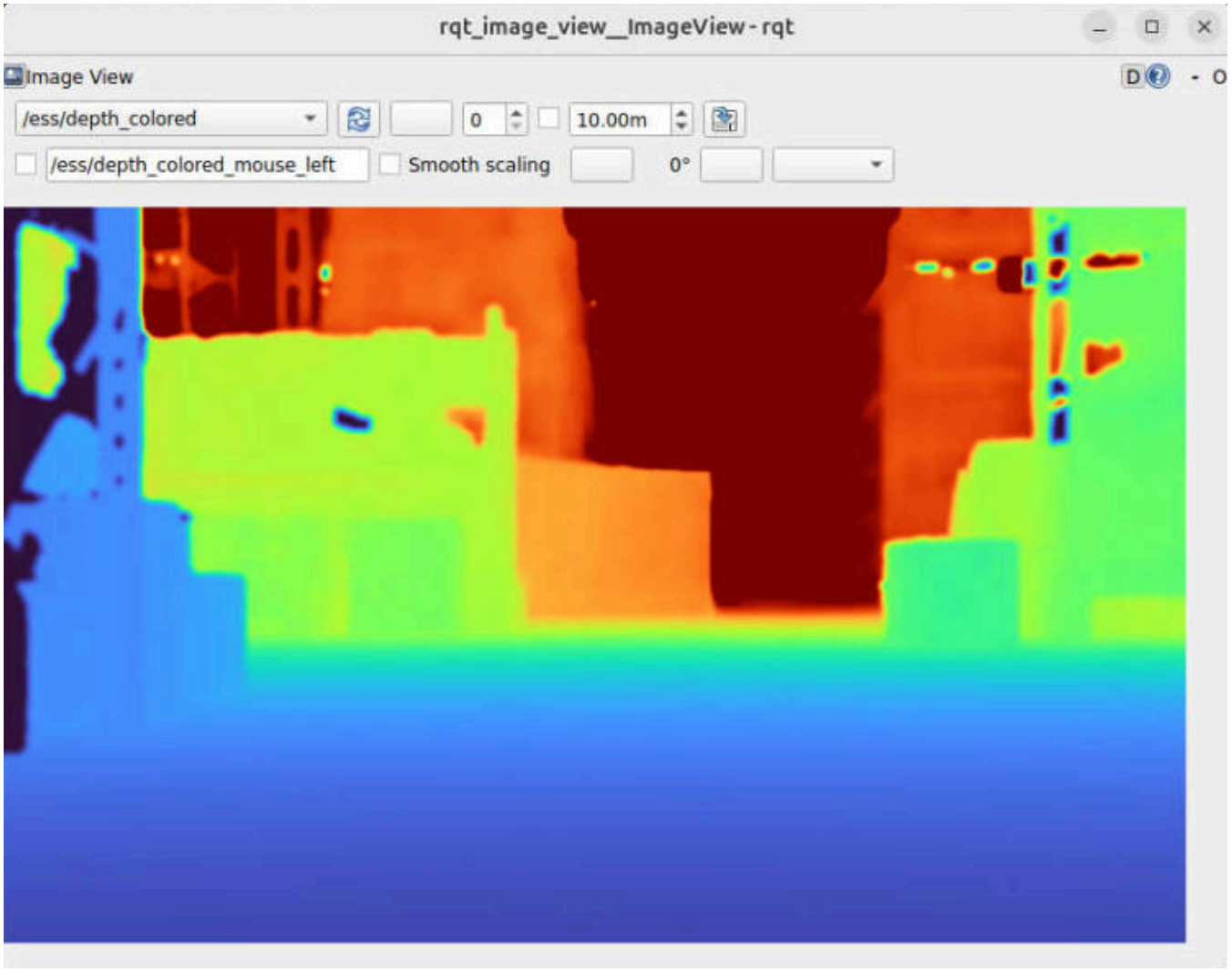
```
ros2 bag play /workspaces/isaac_ros-dev/isaac_ros_assets/isaac_ros_ess/rosbags/ess_rosbag \
  --loop --rate 0.25 --remap \
  /left/camera_info:=/left/camera_info_rect \
  /right/camera_info:=/right/camera_info_rect
```

Terminal C: run the depth segmentation post-processing:

```
python3 /workspaces/isaac_ros-dev/ess_segmentation.py
```

Terminal D: open rqt image view and select `/ess/depth_segmentation` under Topic:

```
ros2 run rqt_image_view rqt_image_view
```


rqt_image_view__ImageView - rqt
Image View
/ess/depth_colored
0
10.00m
/ess/depth_colored_mouse_left
Smooth scaling
0°



